

FIN302

Tutorial 7 Answers

Chapter 1

1.12

- (a) The investor is obligated to sell pounds for 1.3000 when they are worth 1.2900. The gain is $(1.3000 - 1.2900) \times 100,000 = \$1,000$.
- (b) The investor is obligated to sell pounds for 1.3000 when they are worth 1.3200. The loss is $(1.3200 - 1.3000) \times 100,000 = \$2,000$.

1.13

- (a) The trader sells for 50 cents per pound something that is worth 48.20 cents per pound. Gain = $(\$0.5000 - \$0.4820) \times 50,000 = \900 .
- (b) The trader sells for 50 cents per pound something that is worth 51.30 cents per pound. Loss = $(\$0.5130 - \$0.5000) \times 50,000 = \650 .

1.30

Suppose that the forward price for the contract entered into on July 1, 2021, is F_1 and that the forward price for the contract entered into on September 1, 2021, is F_2 with both F_1 and F_2 being measured as dollars per yen. If the value of one Japanese yen (measured in US dollars) is S_T on January 1, 2022, then the value of the first contract (in millions of dollars) at that time is,

$$10(S_T - F_1)$$

while the value of the second contract at that time is:

$$10(F_2 - S_T)$$

The total payoff from the two contracts is therefore,

$$10(S_T - F_1) + 10(F_2 - S_T) = 10(F_2 - F_1)$$

Thus, if the forward price for delivery on January 1, 2022, increased between July 1, 2021, and September 1, 2021, the company will make a profit. (Note that the yen/USD exchange rate is usually expressed as the number of yen per USD not as the number of USD per yen.)

Chapter 2

2.11

There will be a margin call when \$1,000 has been lost from the margin account. This will occur when the price of silver increases by $1,000/5,000 = \$0.20$. The price of silver must therefore rise to \$17.40 per ounce for there to be a margin call. If the margin call is not met, your broker closes out your position.

2.17

There is a margin call if more than \$1,500 is lost on one contract. This happens if the futures price of frozen orange juice falls by more than 10 cents to below 150 cents per pound. \$2,000 can be withdrawn from the margin account if there is a gain on one contract of \$1,000. This will happen if the futures price rises by 6.67 cents to 166.67 cents per pound.

2.21

The clearing house member is required to provide $20 \times \$2,000 = \$40,000$ as initial margin for the new contracts. There is a gain of $(50,200 - 50,000) \times 100 = \$20,000$ on the existing contracts. There is also a loss of $(51,000 - 50,200) \times 20 = \$16,000$ on the new contracts. The member must therefore add

$$40,000 - 20,000 + 16,000 = \$36,000$$

to the margin account.

Chapter 5

5.13

The futures price is $350e^{(0.04-0.03) \times 0.3333} = 351.17$.

5.17

- a) The forward price, F_0 , is given by equation (5.1) as:

$$F_0 = 40e^{0.05 \times 1} = 42.05$$

or \$42.05. The initial value of the forward contract is zero.

- b) The delivery price K in the contract is \$42.05. The value of the contract, f , after six months is given by equation (5.5) as:

$$f = 45 - 42.05e^{-0.05 \times 0.5} = 3.99$$

i.e., it is \$3.99. The forward price is:

$$45e^{0.05 \times 0.5} = 46.14$$

or \$46.14.

5.18

Using equation (5.3), the six month futures price is

$$150e^{(0.07-0.032) \times 0.5} = 152.88$$

or \$152.88.